Future research roadmap: from one verified round to repeated, measured, human allocation

Each study starts from the same unresolved limitation and has an owner, a test that counts as progress, and a date.

UNRESOLVED LIMITATION

Truthful reporting is verified for one allocation round of one synthetic instance, with fixed carbon rates.

- Across repeated rounds, credits may lose their cost (C2 untested over time).
- The same low-value teams can be excluded every round.
- Carbon rates are assumed, not measured, and exact search stops scaling.
- Five author self-plays cannot separate misunderstanding from wanting to win.

Sources: Section 5, Appendices A and D

ECONOMICS

Owner: Mustafa Ayub Khan

Question	Method and data	Possible collaborator	Progress criterion	Milestone
Do fixed-budget credits keep the truth optimal over many rounds, and how should λ and the credit supply be announced?	Multi-round simulation of C2 budgets with carry-over; a pre-announced λ path in the spirit of forward guidance [Hall & Woodford 2026].	DKU research-computing office; FP8 (shared risk-reserve benchmark)	No profitable misreport over 10 rounds; budget never binds	Dec 2026

COMPUTER SCIENCE

Owner: Temur Akhtamjonov, Gihun Lee

Question	Method and data	Possible collaborator	Progress criterion	Milestone
Does the rule scale to many GPU types with measured emissions, without excluding the same teams every round?	Integer program checked against exact search; measured power × grid intensity [Radovanović et al. 2021]; history-based credits [Karma, OSDI 2023].	DKU HPC administrators; FP8	Matches exact search up to 12 teams; 100 teams solved; no team excluded > 3 rounds; DSIC check passes	Spring 2027

BEHAVIORAL SCIENCE

Owner: Gihun Lee

Question	Method and data	Possible collaborator	Progress criterion	Milestone
Do overreports come from misunderstanding the rule or from wanting to win?	Write-a-reason-before-the-result protocol in the Space; classroom sample of 30 or more; preregistered [Li 2017; Cooper & Fang 2008].	COMSCI/ECON 206 class; Prof. Luyao Zhang	Each overreport classified from a written reason; share of truthful reports with 95% interval	Next course offering, 2027

Timeline



Sept 28, 2026
PS2 symposium: one round, one instance

Oct 21, 2026
Final: credit design C1–C3 checked

Dec 2026
Multi-round credit study

2027
GPU types, fairness, classroom sample

2056
Rule adopted by an institution, audited in public